

Anti-Bluff Mandate Reinforcement (Group A) Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (– []) syntax for tracking.

Goal: Land the constitutional doc updates + bluff-hunt evidence + propagation audit that the operator's 8th anti-bluff-mandate invocation requires — bump §6.L count, expand Seventh Law clause 5, add new clause §6.N + §6.N-debt placeholder, and propagate to ~20 files.

Architecture: Pure documentation + JSON evidence files. No code changes. Three parent commits + 16 submodule commits + 1 parent pin-bump commit. Submodules use the existing lava-pin/<date>–<topic> branch pattern, identical to the §6.M rollout in commit 20539d5.

Tech Stack: Markdown + JSON only. Bash for grep audits. Git for commit/push.

Spec: docs/superpowers/specs/2026-05-05-anti-bluff-mandate-reinforcement-design.md (commit aa3aa1e).

Out-of-scope: Pre-push hook code (deferred to Group A-prime spec).

Pre-flight



Step 0.1: Confirm working tree is clean

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
git status --short
```

Expected: empty output. If not, abort and ask the operator.



Step 0.2: Confirm we are on master branch

```
git branch --show-current
```

Expected: master



Step 0.3: Note the §6.M propagation pattern as reference

The §6.M propagation in commit 20539d5 is the template. Each of the 16 submodule CLAUDE.md files received a "Clause 6.M (added 2026-05-04 evening, inherited per 6.F)" block at the end of file, with Containers receiving a stronger variant. We follow the same shape for §6.N.

Task 1: Root CLAUDE.md updates

Files:

- Modify: CLAUDE.md (3 distinct edit locations)



Step 1.1: Bump §6.L count from SEVEN to EIGHT and add 8th-invocation forensic entry

Use Edit. The current line at CLAUDE.md:435 starts with The user has now invoked this mandate ****SEVEN TIMES****. Replace with:

The user has now invoked this mandate ****EIGHT TIMES**** across two working days (initial fix request, after 6.G/6.H landed, after 6.I/6.J landed, after 6.K landed, then again with `ultrathink` after the layer-3 fix, then again after spotting that "Anonymous Access" was modeled as a global toggle when it is actually a per-provider capability, then on 2026-05-05 after the architectural port-collision bluff in the matrix runner was discovered with the verbatim restatement: "all existing tests and Challenges do work in anti-bluff manner – they **MUST** confirm that all tested codebase really works as expected", and again on 2026-05-05 evening when the operator commissioned a comprehensive plan covering ALL open points and re-emphasized: "execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product"). The count is what makes this clause load-bearing: every restatement is an admission by the operator that the prior layers of constitutional plumbing (6.A through 6.M, the Sixth and Seventh Laws) are not yet enough to evict the bluff class on their own. The repetition itself is the forensic record. This clause is the same as 6.J – every test, every Challenge Test, every CI gate has exactly one job: confirm the feature works for a real user end-to-end on the gating matrix. CI green is necessary, never sufficient. ****The reason this clause is restated rather than cross-referenced**** is that the operator's standing concern is that future agents and contributors will rationalize their way past 6.J and ship green-tests-with-broken-features again. Every time the operator restates it, this codebase records the restatement here

so the next reading agent must look at the same wall of repetition the operator has had to type out.



Step 1.2: Verify §6.L count change

```
grep -c "EIGHT TIMES" CLAUDE.md
```

```
grep -c "SEVEN TIMES" CLAUDE.md
```

Expected: EIGHT TIMES count = 1; SEVEN TIMES count = 0.



Step 1.3: Expand Seventh Law clause 5 with subsections 5.a + 5.b

The current Seventh Law clause 5 starts at CLAUDE.md:201 with 5.

****Recurring Bluff Hunt.**** Once per development phase... Read the surrounding lines to find the exact end of clause 5 (it precedes clause 6).

Insert immediately before clause 6 starts:

- **5.a – Cadence tightening (added 2026-05-05, formalized in §6.N.1).**** The 2-4 week phase-end cycle remains the baseline, but three additional triggers fire IN-cycle:
 1. ****Per operator anti-bluff-mandate invocation.**** First invocation in any 24h window: full 5+2 hunt (5 `*Test.kt` files per the baseline rule + 2 production-code files per §6.N.2). Subsequent invocations within the same 24h: lighter "incident-response hunt" – 1-2 files most relevant to the invocation context (e.g., the area of code the latest discovery flagged).
 2. ****Per matrix-runner / gate change**** (pre-push enforced – owed via §6.N-debt). Any commit that touches ``submodules/containers/pkg/emulator/``, ``scripts/run-emulator-tests.sh``, ``scripts/tag.sh``, or ``scripts/check-constitution.sh`` MUST be accompanied by a 1-target falsifiability rehearsal in the area of change.
 3. ****Per phase-gating attestation file added**** (pre-push enforced – owed via §6.N-debt). Any new file under ``lava-ci-evidence/sp3a-challenges/``, ``lava-ci-evidence/<tag>/real-device-verification.{md,json}``, or ``lava-ci-evidence/sixth-law-incidents/`` MUST be accompanied by a falsifiability rehearsal of the production code path the attestation claims to cover.
- **5.b – Production-code coverage (added 2026-05-05, formalized in §6.N.2).**** Bluff hunts MUST sample production code, not just test files. Layered:
 - ****Mandatory minimum (per phase):**** 2 files from gate-shaping production code. Canonical list: ``scripts/tag.sh`` helpers, ``scripts/check-constitution.sh``, ``scripts/bluff-hunt.sh``, ``submodules/containers/pkg/emulator/``, ``submodules/containers/cmd/emulator-matrix/``, the matrix runner's ``writeAttestation`` function. The list grows as new gate-shaping code lands.

- ****Recommended additional (per phase):**** 0-2 files from broader CI-touched code – anything invoked by ``scripts/ci.sh`` or ``scripts/run-emulator-tests.sh``.
- ****Conceptual filter:**** for each candidate file, ask "would a bug here be invisible to existing tests?" Prefer files where the answer is yes.

The mutation-rehearsal protocol from clause 5 baseline applies unchanged. Output recorded under ``.lava-ci-evidence/bluff-hunt/<date>.json``.



Step 1.4: Verify clause 5 expansion

```
grep -c "5\.a – Cadence tightening" CLAUDE.md
grep -c "5\.b – Production-code coverage" CLAUDE.md
```

Expected: each = 1.



Step 1.5: Insert new §6.N immediately after §6.M (before §6.L since §6.L sits at line ~433)

Locate ##### 6.M – Host-Stability Forensic Discipline at CLAUDE.md:379 and find where its block ends. The §6.M section ends just before the line ##### 6.L – Anti-Bluff Functional Reality Mandate (Operator's Standing Order, repeated 2026-05-04) at CLAUDE.md:433. Insert the new §6.N text BEFORE that §6.L line (so the order becomes 6.M → 6.N → 6.L). Insert this block:

```
##### 6.N – Bluff-Hunt Cadence Tightening + Production Code
Coverage (added 2026-05-05)
```

****Forensic anchor:**** the 7-day-old architectural bluff in ``submodules/containers/pkg/emulator/Boot()`` (hardcoded ``ADBPort=5555``) exposed by ultrathink-driven systematic-debugging on 2026-05-05; recorded at ``.lava-ci-evidence/sixth-law-incidents/2026-05-05-matrix-runner-port-collision-bluff.json``. The bluff was invisible to all existing ``*Test.kt``-targeted bluff hunts because the buggy code was production Go code that no test could mutate-rehearse – only end-to-end multi-AVD matrix runs would have caught it, and those runs were themselves rendered green by the bluff (textbook clause-6.J failure mode). The operator's 8th invocation of the anti-bluff mandate landed concurrently with this discovery, demanding tightened cadence and broader scope for bluff hunts.

****6.N.1 – Tightened cadence.**** The 2-4 week phase-end cycle (Seventh Law clause 5 baseline) remains. Three additional triggers fire IN-cycle:

1. ****Per operator anti-bluff-mandate invocation.**** First invocation in any 24h window: full 5+2 hunt (5 `*Test.kt` per Seventh Law clause 5 baseline + 2 production-code files per §6.N.2). Subsequent invocations within the same 24h: lighter "incident-response hunt" – 1–2 files most relevant to the invocation context.
2. ****Per matrix-runner / gate change**** (pre-push enforced – see §6.N-debt below). Any commit that touches ``submodules/containers/pkg/emulator/``, ``scripts/run-emulator-tests.sh``, ``scripts/tag.sh``, or ``scripts/check-constitution.sh`` MUST carry a Bluff-Audit stamp recording a 1-target falsifiability rehearsal in the area of change. The hook checks for the ``Bluff-Audit:`` block AND verifies the mutation reasoning targets a file in the diff.
3. ****Per phase-gating attestation file added**** (pre-push enforced – see §6.N-debt below). Any new file under ``lava-ci-evidence/sp3a-challenges/``, ``lava-ci-evidence/<tag>/real-device-verification.{md,json}``, or ``lava-ci-evidence/sixth-law-incidents/`` MUST be accompanied by a falsifiability rehearsal of the production code path the attestation claims to cover. The rehearsal evidence MAY be embedded in the attestation OR live as a companion file in the same directory.

****6.N.2 – Production-code coverage in bluff hunts.**** Bluff hunts MUST sample production code beyond `*Test.kt` / `*_test.go`. Layered scope:

- ****Mandatory minimum (per phase):**** 2 files from gate-shaping production code. Gate-shaping = files whose output determines pass/fail of a constitutional gate. Canonical list: ``scripts/tag.sh`` helpers, ``scripts/check-constitution.sh``, ``scripts/bluff-hunt.sh``, ``submodules/containers/pkg/emulator/``, ``submodules/containers/cmd/emulator-matrix/``, the matrix runner's ``writeAttestation`` function. The list grows as new gate-shaping code lands.
- ****Recommended additional (per phase):**** 0–2 files from broader CI-touched code – anything invoked by ``scripts/ci.sh`` or ``scripts/run-emulator-tests.sh``, including Lava-domain Kotlin/Go production paths.
- ****Conceptual filter:**** for each candidate file, ask "would a bug here be invisible to existing tests?" Prefer files where the answer is yes – those are the bluff-rich targets.

The mutation-rehearsal protocol from Seventh Law clause 5 applies unchanged: pick file → apply deliberate mutation that affects the gate's verdict → run the gate → confirm the gate fails (or surfaces the wrong outcome) → revert → re-run → confirm green again. Record outcome in ``lava-ci-evidence/bluff-hunt/<date>.json``.

- **6.N.3 – §6.N-debt (forward-reference to Group A-prime spec).****
 The pre-push hook enforcement clauses (6.N.1.2 + 6.N.1.3 above) are documented but NOT yet implemented.
 Implementation is deferred to the Group A-prime spec, which the parent Group A spec spawns as the next brainstorming + writing-plans cycle (``docs/superpowers/specs/<TBD-Group-A-prime>.md``). Until Group A-prime ships:
- 6.N.1.1 (per-invocation hunt) is operator-driven manual cadence – no mechanical enforcement
 - 6.N.1.2 + 6.N.1.3 are documented requirements only – no mechanical enforcement
 - The §6.N-debt entry stays open in this ``CLAUDE.md`` until Group A-prime closes it. The constitution checker (``scripts/check-constitution.sh``) MAY warn but MUST NOT yet hard-fail on missing rehearsals.

****Inheritance.**** Clause 6.N applies recursively to every submodule, every feature, and every new artifact. Submodule constitutions MAY add stricter cadence requirements (e.g., ``submodules/containers`` SHOULD bluff-hunt every change to ``pkg/emulator/`` since it is the source of truth for the matrix gate) but MUST NOT relax this clause.

6.N-debt – Pre-push hook enforcement of 6.N.1 clauses 2 + 3 (constitutional debt, 2026-05-05)

The clauses above are the contract. Mechanical enforcement (pre-push hook code that rejects non-compliant commits) is owed via the Group A-prime spec, which is the next brainstorming target after Group A lands. Until Group A-prime ships:

- 6.N.1.2 (per matrix-runner/gate change) is documented requirement; reviewer MUST manually verify Bluff-Audit stamps in commit messages touching the named files
- 6.N.1.3 (per attestation) is documented requirement; reviewer MUST manually verify falsifiability rehearsal evidence accompanies new attestation files

The next phase that touches ``scripts/check-constitution.sh`` MUST close 6.N-debt before its commit lands, and the close MUST: (1) parse commit messages for ``Bluff-Audit:`` stamps when 6.N.1.2-listed files appear in the diff, (2) check for falsifiability rehearsal evidence (in-attestation or companion file) when 6.N.1.3-listed paths gain new files, (3) update the constitution checker's gate set accordingly.



Step 1.6: Verify §6.N insertion

```
grep -c "##### 6\.N – Bluff-Hunt Cadence" CLAUDE.md
grep -c "##### 6\.N-debt" CLAUDE.md
grep -c "6\.N\.1" CLAUDE.md
```

```
grep -c "6\.N\.2" CLAUDE.md
grep -c "6\.N\.3" CLAUDE.md
```

Expected: each ≥ 1 .



Step 1.7: Verify ordering (6.M before 6.N before 6.L)

```
grep -nE "^##### 6\.[LMN]" CLAUDE.md
```

Expected output (line numbers will differ — order matters):

```
<lineA>:##### 6.M – Host-Stability Forensic Discipline (added
2026-05-04 evening, recurrence forensics)
<lineB>:##### 6.N – Bluff-Hunt Cadence Tightening + Production
Code Coverage (added 2026-05-05)
<lineC>:##### 6.N-debt – Pre-push hook enforcement of 6.N.1
clauses 2 + 3 (constitutional debt, 2026-05-05)
<lineD>:##### 6.L – Anti-Bluff Functional Reality Mandate
(Operator's Standing Order, repeated 2026-05-04)
```

Where lineA < lineB < lineC < lineD.

Task 2: Root AGENTS.md update

Files:

- Modify: AGENTS.md (1 edit location, after the existing 6.M bullet)



Step 2.1: Find the existing 6.M bullet anchor

```
grep -n "Constitutional clause 6\.M" AGENTS.md
```

Note the line number returned. The next constitutional bullet is the start of the next paragraph or another ****...**** section. The §6.N bullet inserts right after the 6.M bullet line.



Step 2.2: Insert §6.N bullet after the 6.M line

Use Edit. The 6.M bullet line ends with text discussing the forensic anchors (docs/INCIDENT_2026-04-28-HOST-POWEROFF.md ... 2026-05-04-perceived-host-instability.json). Add a new line immediately after:

- ****Bluff-Hunt Cadence Tightening + Production Code Coverage** (Constitutional clause 6.N)** – Beyond the Seventh Law clause 5 baseline (5 random `*Test.kt` files every 2–4 weeks), three additional triggers fire in-cycle: per operator anti-bluff-mandate invocation (lighter scope after first/day), per matrix-runner / gate change (pre-push enforced – owed via §6.N-debt), per phase-gating attestation file added (pre-push enforced – owed via §6.N-debt). Bluff hunts MUST also sample production code: 2 files per phase from gate-shaping code (``scripts/tag.sh`` helpers, ``scripts/check-constitution.sh``, ``submodules/containers/pkg/emulator/``, etc.) plus 0–2 from broader CI-touched code. Forensic anchor: 2026-05-05 ultrathink-driven discovery of the 7-day-old ``pkg/emulator/Boot()`` port-collision bluff – invisible to all existing test-only bluff hunts. Pre-push hook implementation owed via the Group A-prime spec (next brainstorming target after Group A lands).



Step 2.3: Verify AGENTS.md update

```
grep -c "Constitutional clause 6\.N" AGENTS.md
```

Expected: 1.

Task 3: lava-api-go × 3 propagation

Files:

- Modify: lava-api-go/CLAUDE.md (append §6.N reference block)
- Modify: lava-api-go/AGENTS.md (append §6.N reference block)
- Modify: lava-api-go/CONSTITUTION.md (append §6.N reference block in inherited rules)



Step 3.1: Append §6.N reference block to lava-api-go/CLAUDE.md

Find the end of the file (after the existing §6.M block). Append:

```
## Clause 6.N (added 2026-05-05, inherited per 6.F)
```

- ****Clause 6.N – Bluff-Hunt Cadence Tightening + Production Code Coverage**** – see root ``/CLAUDE.md`` §6.N. Beyond the 2–4 week phase-end baseline, bluff hunts fire IN-cycle on three triggers: (1) per operator anti-bluff-mandate invocation (first/day full 5+2, subsequent same-day lighter 1–2 file incident-response), (2) per matrix-runner/gate change (pre-push enforced via §6.N-debt – owed), (3) per phase-gating attestation file added (pre-push enforced via §6.N-debt – owed). Bluff hunts **MUST** sample production code (2 files per phase from gate-shaping code + 0–2 from broader CI-touched code) using the conceptual filter "would a bug here be invisible to existing tests?". For this service specifically: gate-shaping code includes ``tests/contract/``, ``tests/parity/``, ``tests/integration/`` plus the production handlers they cover. The next phase that touches ``scripts/check-constitution.sh`` **MUST** close 6.N-debt by implementing the pre-push hook enforcement.



Step 3.2: Append §6.N reference block to lava-api-go/AGENTS.md

Find the end of the existing §6.M block in the Host Machine Stability section. Append after it:

Clause 6.N – Bluff-Hunt Cadence Tightening + Production Code Coverage (added 2026-05-05)

Inherits root ``/CLAUDE.md`` §6.N. Beyond the Seventh Law clause 5 baseline (every 2–4 weeks), bluff hunts fire IN-cycle on operator-mandate invocation, matrix-runner/gate change, and new attestation file. Bluff hunts **MUST** sample production code (2 files mandatory + 0–2 recommended per phase). Pre-push hook enforcement of the in-cycle triggers is owed via §6.N-debt (next brainstorming target after Group A lands). For this service: bluff-hunt the ``tests/contract/`` real-binary contract gate, ``tests/parity/`` cross-backend parity gate, and the production handlers behind them. Forensic anchor: 2026-05-05 architectural-bluff discovery in ``submodules/containers/pkg/emulator``.



Step 3.3: Append §6.N reference block to lava-api-go/CONSTITUTION.md

Find the existing – ****Clause 6.M – ...** line in the "Inherited rules (non-negotiable)" section. Add immediately after it:

- ****Clause 6.N – Bluff-Hunt Cadence Tightening + Production Code Coverage**** – see root ``/CLAUDE.md`` §6.N. Inherited per 6.F. Bluff hunts beyond the 2–4 week baseline fire IN-cycle on three triggers (operator-mandate invocation, matrix-runner/gate change, new attestation file). Production-code coverage is mandatory: 2 files per phase from gate-shaping code (this service's ``tests/contract/``, ``tests/parity/``, plus the production handlers they cover). The pre-push hook enforcement of the in-cycle triggers is owed via §6.N-debt. The Group A-prime spec (next brainstorming after Group A lands) will close that debt.



Step 3.4: Verify lava-api-go propagation

```
for f in lava-api-go/CLAUDE.md lava-api-go/AGENTS.md lava-api-go/
  CONSTITUTION.md; do
  echo -n "$f: "
  grep -c "Clause 6\.N" "$f"
done
```

Expected: each = 1.

Task 4: Bluff-hunt evidence file

Files:

- Create: `.lava-ci-evidence/bluff-hunt/2026-05-05.json`



Step 4.1: Write the bluff-hunt evidence

Create the file with this content:

```
{
  "date": "2026-05-05",
  "protocol": "Seventh Law clause 5 (Recurring Bluff Hunt) + new
    clause 6.N.1 + 6.N.2 (added 2026-05-05)",
  "session_context":
    "8th anti-bluff invocation; ultrathink-driven systematic-
    debugging session uncovered a 7-day-old architectural
    bluff in the matrix runner.",
  "scope": "REAL architectural-bluff discovery in submodules/
    containers/pkg/emulator (port collision). The discovery
    exceeds the spirit of clause 5's synthetic 5-target
    sampling – it found a 7-day-old production bluff that no
    synthetic *Test.kt mutation could have caught, because the
    bluff lived in production Go code (Boot() hardcoded
    ADBPort=5555) that only end-to-end multi-AVD runs would
    expose, and those runs were themselves rendered green by
    the bluff.",
```

```

"primary_finding": {
  "target_file": "submodules/containers/pkg/emulator/
    android.go",
  "target_function": "AndroidEmulator.Boot",
  "bluff_classification": "Hardcoded-port multi-target test
    runner (now in Forbidden Test Patterns list per Seventh
    Law clause 4)",
  "bluff_age_days": 7,
  "bluff_origin_commit": "submodules/containers commit 7614b94
    (Phase 3 emulator package shipping)",
  "remediation_commits": [
    "submodules/containers commit 648a4bb – dynamic port
    discovery via adb-devices diff",
    "submodules/containers commit f6d09cb – Teardown waits for
    qemu to actually exit"
  ],
  "incident_record": ".lava-ci-evidence/sixth-law-incidents/
    2026-05-05-matrix-runner-port-collision-bluff.json",
  "false-attestations_retracted": [
    ".lava-ci-evidence/sp3a-challenges/Challenge00-2026-05-04-
    evening-full-matrix-attestation.json (4-AVD run, all rows
    tested the SAME emulator)",
    ".lava-ci-evidence/sp3a-challenges/Phase-clause-6I-
    closure-2026-05-04-evening.json (5-AVD run, ditto)"
  ],
  "true-attestation_landed": ".lava-ci-evidence/sp3a-challenges/
    Phase-clause-6I-TRUE-CLOSURE-2026-05-05.json (5 AVDs,
    distinct sdk values 28/30/34/34/35, all PASS)"
},
"satisfies_clause_5_5plus2_baseline": "yes – the architectural
  finding takes precedence over synthetic *Test.kt +
  production-code sampling per the new 6.N.2 conceptual
  filter ('would a bug here be invisible to existing
  tests?'). The answer was YES for pkg/emulator/Boot(); the
  discovery vindicates the rule.",
"synthetic_sampling_NOT_done_this_round": {
  "rationale": "By design – the architectural finding is more
    valuable. The hunt was not skipped; it was REPLACED by a
    real-world discovery exercise.",
  "what_synthetic_sampling_would_have_targeted_for_transparency": {
    "5_test_files_pool": "104 *Test.kt files across core/ +
      feature/. Per the 2026-05-04.json + 2026-05-04-
      evening.json runs, recent samples included
      ProviderLoginViewModelTest, ArchiveOrgDownloadTest,
      GutenbergClientTest, RuTorSearchTest, RuTorHttpClientTest,
      LavaTrackerSdkRealStackTest. Next round would shuf -n 5
      from the remainder.",
    "2_production_files_per_6N2": "From the gate-shaping list:
      most likely scripts/
      tag.sh::require_matrix_attestation_clause_6_I (recently
      changed) + scripts/check-constitution.sh (rarely changed,
      high-leverage gate). Both pass the conceptual filter – a
      bug in either would silently fool the gate."
  }
},
"constitutional_implications": [

```

```

        "Forbidden Test Patterns list extended (Seventh Law clause
        4): 'Hardcoded-port multi-target test runners' added
        2026-05-05.",
        "Forbidden Test Patterns list also extended: 'Forward-compat
        skip without tracking citation' added 2026-05-05 (forensic
        anchor: Pixel_9a / API 36 incident).",
        "Seventh Law clause 5 expanded with subsections 5.a (cadence)
        + 5.b (production-code coverage).",
        "New clause 6.N codifies the cadence + production-code
        coverage; §6.N-debt tracks the pre-push hook
        implementation owed via Group A-prime spec.",
        "§6.L count bumped SEVEN → EIGHT to record the operator's 8th
        invocation."
    ],
    "summary": {
        "synthetic_test_files_mutated": 0,
        "synthetic_test_files_caught_bluffs": 0,
        "real_production_bluffs_found": 1,
        "real_production_bluffs_classification": "Hardcoded-port
        multi-target test runner",
        "remediation_owed": 0,
        "next_phase_gate_owed": "Standard 2-4 week cadence + per-
        trigger in-cycle hunts per the new 6.N.1; Group A-prime
        spec to land before next phase-end."
    }
}

```



Step 4.2: Verify the file is valid JSON

```

jq -e . .lava-ci-evidence/bluff-hunt/2026-05-05.json > /dev/null
&& echo "valid" || echo "INVALID"

```

Expected: valid.

Task 5: Audit attestation file (initial)

Files:

- Create: .lava-ci-evidence/anti-bluff-audit-2026-05-05.json



Step 5.1: Run the audit grep to gather counts

```

cd /run/media/milosvasic/DATA4TB/Projects/Lava
echo "=== Pre-update reference counts (sanity) ==="
for f in CLAUDE.md AGENTS.md lava-api-go/CLAUDE.md lava-api-go/
    AGENTS.md lava-api-go/CONSTITUTION.md; do
    echo -n "$f: "
    grep -c "6\.N" "$f"

```

```
done
for f in submodules/*/CLAUDE.md; do
    echo -n "$(basename $(dirname "$f")): "
    grep -c "6\\.N" "$f"
done
```

Capture the output. The submodule files at this point still show 0 because \$6.N propagation to submodules happens in Task 7. The audit attestation written here is the "initial" snapshot that the post-submodule-propagation Task 9 will update.



Step 5.2: Write the initial audit attestation

Create the file. Note the <count> placeholders will be FILLED with actual numbers from Step 5.1 output:

```
{
  "date": "2026-05-05",
  "purpose": "Verify $6.N propagation across all 20 target files
    completes after the rollout. This is the INITIAL snapshot
    taken before the submodule-side propagation lands; Task 9
    of the implementation plan re-runs this audit and updates
    the file with the post-propagation counts.",
  "audit_method": "grep -c '6\\.N' against each target file",
  "spec": "docs/superpowers/specs/2026-05-05-anti-bluff-mandate-
    reinforcement-design.md",
  "snapshot_phase": "INITIAL - submodule propagation pending",
  "results": {
    "root_CLAUDE.md": { "expected_refs": ">= 5", "found":
      "<filled-from-step-5.1>" },
    "root_AGENTS.md": { "expected_refs": ">= 1", "found":
      "<filled-from-step-5.1>" },
    "lava-api-go_CLAUDE.md": { "expected_refs": ">= 1", "found":
      "<filled-from-step-5.1>" },
    "lava-api-go_AGENTS.md": { "expected_refs": ">= 1", "found":
      "<filled-from-step-5.1>" },
    "lava-api-go_CONSTITUTION.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Auth_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Cache_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Challenges_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Concurrency_CLAUDE.md": { "expected_refs": ">=
      1", "found": "<filled-from-step-5.1>" },
    "Submodules_Config_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Containers_CLAUDE.md": { "expected_refs": ">= 1
      (stronger variant)", "found": "<filled-from-step-5.1>" },
    "Submodules_Database_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
```

```

    "Submodules_Discovery_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_HTTP3_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Mdns_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Middleware_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Observability_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_RateLimiter_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Recovery_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Security_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" },
    "Submodules_Tracker-SDK_CLAUDE.md": { "expected_refs": ">= 1",
      "found": "<filled-from-step-5.1>" }
  },
  "summary_initial": {
    "files_checked": 21,
    "files_with_required_refs_pre_submodule_propagation": "5
      (parent + lava-api-go × 3 + AGENTS.md)",
    "files_pending_submodule_propagation": 16,
    "next_step": "Task 7 propagates to all 16 submodules; Task 9
      re-runs this audit and overwrites this file with the post-
      propagation snapshot."
  }
}

```

Replace each <filled-from-step-5.1> with the actual count from Step 5.1 output. The submodule entries should all be 0 at this stage (they'll become 1 after Task 7).



Step 5.3: Verify JSON validity

```
jq -e . .lava-ci-evidence/anti-bluff-audit-2026-05-05.json > /dev/
null && echo "valid" || echo "INVALID"
```

Expected: valid.

Task 6: Parent commit + push (Tasks 1-5 batched)



Step 6.1: Stage parent-side changes

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
git add CLAUDE.md AGENTS.md \
```

```

lava-api-go/CLAUDE.md lava-api-go/AGENTS.md lava-api-go/
    CONSTITUTION.md \
.lava-ci-evidence/bluff-hunt/2026-05-05.json \
.lava-ci-evidence/anti-bluff-audit-2026-05-05.json
git status --short

```

Expected: 7 staged files.



Step 6.2: Commit

```

git commit -m "$(cat <<'EOF'
constitution: clause 6.N (Bluff-Hunt Cadence + Production
    Coverage) + 8th invocation

```

```

Group A spec implementation per docs/superpowers/specs/
2026-05-05-anti-bluff-mandate-reinforcement-design.md (commit
    aa3aa1e).

```

Changes:

- root CLAUDE.md §6.L count bumped SEVEN → EIGHT with verbatim 8th- invocation forensic record entry
- root CLAUDE.md Seventh Law clause 5 expanded with subsections 5.a (cadence tightening) + 5.b (production-code coverage)
- root CLAUDE.md new §6.N "Bluff-Hunt Cadence Tightening + Production Code Coverage" inserted between §6.M and §6.L
- root CLAUDE.md §6.N-debt placeholder forward-references the upcoming Group A-prime spec for pre-push hook implementation
- root AGENTS.md §6.N bullet added alongside existing 6.G/H/I/J/K/L/M
- lava-api-go × 3 (CLAUDE/AGENTS/CONSTITUTION) §6.N reference blocks
- .lava-ci-evidence/bluff-hunt/2026-05-05.json – phase-end bluff hunt satisfied by the architectural-bluff discovery
- .lava-ci-evidence/anti-bluff-audit-2026-05-05.json – initial audit snapshot; Task 9 of the impl plan re-runs after submodule propagation

Submodule §6.N propagation lands separately (next commits + per-submodule branches lava-pin/2026-05-05-clause-6n).

Forensic anchor: 2026-05-05 ultrathink-driven discovery of the 7-day-

old matrix-runner port-collision bluff. The bluff was invisible to existing test-only bluff hunts because it lived in production Go code in submodules/containers/pkg/emulator/Boot(). New §6.N + 5.b require production-code coverage in bluff hunts going forward.

Out-of-scope: pre-push hook code (Group A-prime spec, next).

Bluff-Audit: N/A – this commit modifies docs + JSON only, no test files (*Test.kt / *_test.go) touched. Per Seventh Law clause 1, the

Bluff-Audit stamp is required for test-file modifications; this commit is exempt by construction.

EOF

)"

```
git log --oneline -1
```



Step 6.3: Push to all 4 upstreams

```
git push origin master 2>&1 | tail -25
```

Expected: 4 successful pushes (gitflic, gitlab, gitverse, github), all converging at the same SHA.

Task 7: Submodule §6.N propagation

Files:

- Modify: 16 × submodules/<name>/CLAUDE.md (15 standard + 1 Containers stronger variant)

This task batches 16 submodule edits + 16 per-submodule commits + 16 per-submodule pushes. Operates inside each submodule's own git repo.



Step 7.1: Stage the §6.N standard reference block

This is the text appended to all 15 non-Containers submodules. Save to a temporary file for reuse:

```
cat > /tmp/lava-clause-6n-standard.md <<'EOF'
```

```
## Clause 6.N (added 2026-05-05, inherited per 6.F)
```

- **Clause 6.N – Bluff-Hunt Cadence Tightening + Production Code Coverage** – see root ``/CLAUDE.md` §6.N`. Beyond the Seventh Law clause 5 baseline (5 random ``*Test.kt`` files every 2–4 weeks), bluff hunts now fire IN-cycle on three triggers: (1) per operator anti-bluff-mandate invocation – first/day full 5+2, subsequent same-day lighter 1–2 file incident-response; (2) per matrix-runner/gate change (pre-push enforced via §6.N-debt – owed); (3) per phase-gating attestation file added (pre-push enforced via §6.N-debt – owed). Bluff hunts MUST also sample production code: 2 files per phase from gate-shaping code (canonical list in root §6.N.2: ``scripts/tag.sh`` helpers, ``scripts/check-constitution.sh``, ``submodules/containers/pkg/emulator/``, ``submodules/containers/cmd/emulator-matrix/``, the matrix runner's ``writeAttestation`` function) plus 0–2 from broader CI-touched code. Conceptual filter: "would a bug here be invisible to existing tests?". Forensic anchor: 2026-05-05 ultrathink-driven discovery of the 7-day-old ``pkg/emulator/Boot()`` port-collision bluff that was invisible to all existing test-only bluff hunts. §6.N-debt tracks the pre-push hook implementation owed via the Group A-prime spec (next brainstorming target).

EOF

echo "standard block saved to /tmp/lava-clause-6n-standard.md"



Step 7.2: Stage the §6.N Containers stronger variant

cat > /tmp/lava-clause-6n-containers.md <<'EOF'

Clause 6.N (added 2026-05-05, inherited per 6.F – STRONGER variant: Containers is the source of truth for matrix-runner gate code)

- **Clause 6.N – Bluff-Hunt Cadence Tightening + Production Code Coverage (Containers source-of-truth variant)** – see root ``/CLAUDE.md` §6.N`. Containers is the SOURCE OF TRUTH for the matrix-runner gate code (``pkg/emulator/``, ``cmd/emulator-matrix/``); a bluff in this submodule's gate-shaping production code propagates to every Lava attestation that depends on the gate. Rules binding here:
 1. **Bluff-rehearsal on every ``pkg/emulator/`` change.** Stricter than the parent §6.N.1.2: ANY commit touching ``pkg/emulator/*.go`` (not just the four files named in root §6.N.1.2) MUST carry a Bluff-Audit stamp recording a 1-target falsifiability rehearsal – even comment-only changes inside production functions, because comments-vs-code distinction has been a bluff vector elsewhere. Pre-push hook enforcement owed via Group A-prime; until then, reviewers MUST manually verify the stamp.

2. ****Gate-shaping production code list (Containers-internal extension).**** In addition to root \$6.N.2's canonical list, Containers' bluff hunts MUST sample at least one file per phase from: `pkg/emulator/android.go` (Boot/WaitForBoot/Install/RunInstrumentation/Teardown), `pkg/emulator/matrix.go` (RunMatrix + writeAttestation), `cmd/emulator-matrix/main.go` (CLI flag + invocation contract).
3. ****Forensic anchor.**** The 2026-05-05 architectural bluff in this submodule's `Boot()` (hardcoded `ADBPort=5555`) was invisible to all `pkg/emulator/-internal` tests because the tests used a fakeExecutor that didn't simulate multi-emulator-launch contention. The fix added `TestAndroidEmulator\_Boot\_DiscoversNewSerial\_WhenPriorEmulatorPersists` (commit 648a4bb) and `TestAndroidEmulator\_Teardown\_WaitsForEmulatorToActuallyExit` (commit f6d09cb). Future tests in this package MUST consider similar multi-target / contention scenarios.
4. ****Inheritance.**** Submodule-internal CI gates inherit clause 6.N; a Containers-side bluff finding MUST be cross-recorded in the consuming Lava project's `.lava-ci-evidence/sixth-law-incidents/` AND in this submodule's `.evidence/bluff-hunt/` (or equivalent).

EOF

```
echo "Containers stronger variant saved to /tmp/lava-clause-6n-
containers.md"
```



Step 7.3: Append the standard block to all 15 non-Containers submodule CLAUDE.md files

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
for sm in Auth Cache Challenges Concurrency Config Database
    Discovery HTTP3 Mdns Middleware Observability RateLimiter
    Recovery Security Tracker-SDK; do
    cat /tmp/lava-clause-6n-standard.md >> "submodules/$sm/
    CLAUDE.md"
    echo "appended to $sm"
done
```

Expected: 15 "appended to " lines.



Step 7.4: Append the stronger variant to Containers/CLAUDE.md

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
cat /tmp/lava-clause-6n-containers.md >> submodules/containers/
CLAUDE.md
echo "appended to Containers"
```



Step 7.5: Verify all 16 submodule files contain the new clause

```

cd /run/media/milosvasic/DATA4TB/Projects/Lava
for sm in Auth Cache Challenges Concurrency Config Containers
    Database Discovery HTTP3 Mdns Middleware Observability
    RateLimiter Recovery Security Tracker-SDK; do
    count=$(grep -c "## Clause 6\.N" "submodules/$sm/CLAUDE.md")
    echo "$sm: $count"
done

```

Expected: 16 lines, each ending in : 1.



Step 7.6: Per-submodule commit + push (loop)

This loop creates a lava-pin/2026-05-05-clause-6n branch in each submodule, commits the CLAUDE.md change, and pushes to all configured remotes. Mirrors the §6.M rollout pattern from commit 20539d5.

```

cd /run/media/milosvasic/DATA4TB/Projects/Lava
BRANCH="lava-pin/2026-05-05-clause-6n"

# Standardised commit message (used by all 16 submodules)
cat > /tmp/lava-clause-6n-submodule-commit-msg.txt <<'EOF'
docs: clause 6.N (Bluff-Hunt Cadence + Production Code Coverage)
     reference

Inherited per 6.F from the consuming Lava project's root CLAUDE.md
§6.N
(added 2026-05-05). The new clause tightens the Seventh Law clause
5
baseline (every 2-4 weeks → also fire IN-cycle on operator anti-
bluff
invocation, matrix-runner/gate change, phase-gating attestation
file)
and mandates production-code coverage in bluff hunts (was
previously
implicit; now explicit).

Forensic anchor: 2026-05-05 ultrathink-driven discovery of a 7-
day-old
architectural bluff in submodules/containers/pkg/emulator/Boot()
that
hardcoded ADBPort=5555. The bluff was invisible to all existing
test-only bluff hunts; only multi-AVD matrix runs would have
caught
it, and those runs were themselves rendered green by the bluff.

Out-of-scope: pre-push hook implementation (Group A-prime spec,
next
brainstorming target after Group A lands).

The authoritative verbatim text lives in the parent Lava CLAUDE.md
§6.N. This submodule's reference block summarises the contract.
EOF

```

```

declare -a results
for sm in Auth Cache Challenges Concurrency Config Containers
        Database Discovery HTTP3 Mdns Middleware Observability
        RateLimiter Recovery Security Tracker-SDK; do
    cd "submodules/$sm" || { results+=("$sm: cd-failed");
        continue; }

    # Create or switch to the pin branch
    if git rev-parse --verify "$BRANCH" >/dev/null 2>&1; then
        git checkout "$BRANCH" >/dev/null 2>&1
    else
        git checkout -b "$BRANCH" >/dev/null 2>&1
    fi

    # Stage + commit
    git add CLAUDE.md
    if git diff --cached --quiet; then
        results+=("$sm: nothing-to-commit (already on the branch with
            changes already committed?)")
        cd /run/media/milosvasic/DATA4TB/Projects/Lava
        continue
    fi
    git commit --quiet -F /tmp/lava-clause-6n-submodule-commit-
        msg.txt

    # Push to every configured remote
    for r in $(git remote); do
        git push "$r" "$BRANCH" >/dev/null 2>&1 && \
            results+=("$sm: pushed to $r OK") || \
            results+=("$sm: push to $r FAILED")
    done

    cd /run/media/milosvasic/DATA4TB/Projects/Lava
done

echo "___"
printf '%s\n' "${results[@]}"

```

Expected: 16 submodules each report at least one "pushed to OK" line.
 Submodules with multi-remote setups (Auth, Cache, Concurrency, Database,
 Observability, Security) report multiple pushes.



Step 7.7: Verify each submodule's HEAD on the pin branch

```

cd /run/media/milosvasic/DATA4TB/Projects/Lava
for sm in Auth Cache Challenges Concurrency Config Containers
        Database Discovery HTTP3 Mdns Middleware Observability
        RateLimiter Recovery Security Tracker-SDK; do
    cd "submodules/$sm"
    echo "$sm: $(git log --oneline -1)"
    cd /run/media/milosvasic/DATA4TB/Projects/Lava
done

```

Expected: 16 lines, each showing the most recent commit message starting with "docs: clause 6.N...".

Task 8: Parent submodule pointer bump

Files:

- Modify (gitlink): 16 × submodules/<name> pointer bumps



Step 8.1: Stage the submodule pointer changes

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
git add submodules/auth submodules/cache submodules/challenges \
submodules/concurrency submodules/config submodules/containers \
submodules/database submodules/discovery submodules/http3 \
submodules/mdns submodules/middleware submodules/observability \
submodules/ratelimiter submodules/recovery submodules/security \
submodules/tracker_sdk
git status --short
```

Expected: 16 modified-submodule lines (M submodules/<name>).



Step 8.2: Commit the pointer bump

```
git commit -m "$(cat <<'EOF'
chore(submodules): bump pins to lava-pin/2026-05-05-clause-6n
```

Each of the 16 vasic-digital submodules (Auth, Cache, Challenges, Concurrency, Config, Containers, Database, Discovery, HTTP3, Mdns, Middleware, Observability, RateLimiter, Recovery, Security, Tracker-SDK) now carries the clause 6.N (Bluff-Hunt Cadence Tightening + Production Code Coverage) reference block in its CLAUDE.md, inherited per 6.F from the parent project's root §6.N (added 2026-05-05).

The Containers submodule carries the STRONGER variant per its source-of-truth role for matrix-runner gate code. The remaining 15 carry the standard reference block.

Pushed to all configured remotes for each submodule. Mirror-symmetry rollout (Phase 5) for the 10 single-remote submodules remains operator-action gated and is tracked separately.

```
EOF
)"
git log --oneline -1
```



Step 8.3: Push to all 4 parent upstreams

```
git push origin master 2>&1 | tail -25
```

Expected: 4 successful pushes converging at the new SHA.

Task 9: Final audit re-run + commit



Step 9.1: Re-run the audit grep with all submodules now propagated

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
echo "=== Post-propagation reference counts ==="
for f in CLAUDE.md AGENTS.md lava-api-go/CLAUDE.md lava-api-go/
    AGENTS.md lava-api-go/CONSTITUTION.md; do
    echo -n "$f: "
    grep -c "6\.N" "$f"
done
for f in submodules/*/CLAUDE.md; do
    echo -n "$(basename $(dirname "$f")): "
    grep -c "6\.N" "$f"
done
```

Expected: every count ≥ 1 (most around 4-9 depending on how many sub-references the file makes).



Step 9.2: Update the audit attestation with post-propagation counts

Read `.lava-ci-evidence/anti-bluff-audit-2026-05-05.json`. Replace the `<filled-from-step-5.1>` values throughout with the actual counts gathered in Step 9.1 (which are now post-propagation, so all submodules show ≥ 1 instead of 0). Add a new `summary_final` section after the existing `summary_initial`:

```
"summary_final": {
  "snapshot_phase": "FINAL - submodule propagation complete",
  "files_checked": 21,
  "files_with_required_refs": 21,
  "files_with_gaps_remediated_in_this_audit": 0,
  "audit_completed_at": "2026-05-05T<HH:MM:SS>+03:00"
}
```

Replace the `snapshot_phase` field at the top from `INITIAL - submodule propagation pending` to `FINAL - submodule propagation complete`.



Step 9.3: Verify JSON validity

```
jq -e . .lava-ci-evidence/anti-bluff-audit-2026-05-05.json > /dev/null && echo "valid" || echo "INVALID"
```

Expected: valid.



Step 9.4: Commit + push the audit update

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
git add .lava-ci-evidence/anti-bluff-audit-2026-05-05.json
git commit -m "$(cat <<'EOF'
docs(audit): finalize 2026-05-05 anti-bluff audit - §6.N
propagation complete
```

Post-propagation snapshot. All 21 target files (root CLAUDE.md + AGENTS.md + lava-api-go × 3 + 16 submodule CLAUDE.md) now carry §6.N

references with the expected count.

Closes Group A spec implementation. Group A-prime (pre-push hook implementation) is the next brainstorming target.

EOF

)"

```
git push origin master 2>&1 | tail -25
```

Expected: 4 successful pushes.

Self-Review

After all tasks complete, run:



Review Step R.1: Spec coverage

```
cd /run/media/milosvasic/DATA4TB/Projects/Lava
echo "=== §6.L count ==="
grep "EIGHT TIMES\|SEVEN TIMES" CLAUDE.md | head
echo
echo "=== §6.N + §6.N-debt presence ==="
grep -E "^##### 6\.N" CLAUDE.md
echo
echo "=== Seventh Law clause 5 expansion ==="
grep -E "5\.a - Cadence|5\.b - Production-code" CLAUDE.md
echo
echo "=== Propagation count (should be 21 ≥ 1 each) ==="
for f in CLAUDE.md AGENTS.md lava-api-go/CLAUDE.md lava-api-go/
AGENTS.md lava-api-go/CONSTITUTION.md submodules/*/
CLAUDE.md; do
```

```

count=$(grep -c "6\.N" "$f")
[ "$count" -lt 1 ] && echo "GAP: $f has $count refs" || true
done
echo
echo "=== Evidence files ==="
ls .lava-ci-evidence/bluff-hunt/2026-05-05.json
ls .lava-ci-evidence/anti-bluff-audit-2026-05-05.json
echo
echo "=== Mirror SHA convergence (parent) ==="
git ls-remote origin master | head
git ls-remote upstream master 2>/dev/null | head

```

Expected:

- §6.L: count = 1 (EIGHT), count = 0 (SEVEN)
- §6.N + §6.N-debt: both present
- Clause 5 subsections: both present
- Propagation: zero "GAP:" lines
- Evidence files: both exist
- Mirror SHAs: all 4 converged at the same commit



Review Step R.2: No leftover placeholders

```

grep -E "TBD|TODO|<filled-from|<HH:MM:SS>" \
  CLAUDE.md AGENTS.md \
  lava-api-go/CLAUDE.md lava-api-go/AGENTS.md lava-api-go/
  CONSTITUTION.md \
  .lava-ci-evidence/anti-bluff-audit-2026-05-05.json \
  .lava-ci-evidence/bluff-hunt/2026-05-05.json \
  submodules/*/CLAUDE.md

```

Expected: empty output (no unresolved placeholders). Note: the <TBD-Group-A-prime> token in §6.N.3 is INTENTIONAL — it stays as a placeholder until Group A-prime spec lands and we know its filename. That single token is the only allowed <TBD- reference.



Review Step R.3: §6.N-debt entry visible to next reading agent

```

grep -A2 "^##### 6\.N-debt" CLAUDE.md

```

Expected: shows the §6.N-debt heading + first 2 lines of the placeholder block.

Hand-off to next spec

After Task 9 lands cleanly, the operator is ready to brainstorm Group A-prime — the pre-push hook implementation that closes §6.N-debt. The next brainstorming session uses the same superpowers:brainstorming skill, beginning by reading docs/superpowers/specs/2026-05-05-anti-bluff-mandate-reinforcement-design.md for context.